Direction Instability Predicts Cross-Cell-Line Drug Mechanism Transport in LINCS L1000

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Abstract

A drug's mechanism of action may or may not generalize across cell types. We adapt a geometric measure—direction instability, the mean pairwise angular deviation of perturbation signature vectors—to quantify whether a drug's transcriptomic effect is conserved across biological contexts. Applied to 8,949 compounds tested in ≥ 5 cell lines from the LINCS L1000 Connectivity Map (978 landmark genes, 167,266 signatures), direction instability separates mechanism-conserving drugs from context-dependent ones (population-null $p < 0.01$ for 78 drugs). Direction instability anticorrelates with gene-level Jaccard overlap of top perturbed genes (Spearman $\rho = -0.79$), confirming that directional change reflects biological mechanism change. Within the HDAC inhibitor class, pan-isoform inhibitors transport across cell lines (mean instability 0.62, $n = 8$) while isoform-selective inhibitors do not (mean instability 0.96, $n = 5$), demonstrating that target breadth—not drug class membership—determines cross-context generalizability. Genetic triangulation against 154,993 shRNA knockdown signatures showed a significant pre-registered global correlation ($\rho = -0.18$, $p = 1.1 \times 10^{-16}$) that fell below the pre-registered effect-size threshold; exploratory within-family stratification—which controls for the confound that different target classes have inherently different drug–knockdown cosine baselines—revealed stronger effects ($|\rho| = 0.32$–0.84 across five families), a result that requires independent replication.

1 Introduction

Pharmacological perturbation experiments increasingly rely on profiling drug effects across panels of cell lines to identify compounds with robust mechanisms of action (Subramanian et al., 2017). The LINCS L1000 Connectivity Map measures 978 landmark gene expression changes for ∼20,000 compounds across up to 70 cell lines, producing high-dimensional signature vectors that characterize each drug's transcriptomic effect in each context (Subramanian et al., 2017).

A fundamental question arises: when does a drug's effect *transport* from one cell type to another? Two drugs may share a mechanism-of-action label yet differ in how consistently they perturb gene expression across contexts. A pan-HDAC inhibitor like vorinostat affects chromatin remodeling in every cell, while an HDAC8-selective inhibitor like PCI-34051 only produces a strong transcriptomic signature where HDAC8 is functionally relevant.

We formalize this distinction using **direction instability**: the mean pairwise angular deviation of a drug's unit-normalized signature vectors across cell lines. This metric is adapted from the neural bracket norm (Tower, 2026), which quantifies how much a learned representation's direction rotates across evidence levels. Here, the analogy maps evidence quartiles to cell lines, and neural activation vectors to drug perturbation signatures.

Direction instability decomposes the perturbation effect into two independent components. The *direction* captures which genes are perturbed and in what relative proportions. The *magnitude* captures the overall effect size. A drug with low direction instability but high magnitude variation has a conserved qualitative mechanism with context-dependent penetrance—a biologically meaningful and pharmacologically useful category.

We apply this framework to 8,949 drugs with ≥ 5 cell lines in LINCS L1000. The scorer and validation labels were frozen under a pre-registered commit SHA before any real data were analyzed.

2 Methods

2.1 Data

Perturbation signatures were drawn from the LINCS L1000 Phase I dataset (GEO accession GSE92742), using Level 5 replicate-collapsed z-scores across 978 landmark genes (Subramanian et al., 2017). Metadata (signature identifiers, perturbation types, cell line annotations) were downloaded from GEO and filtered to small-molecule perturbagens (`pert_type = trt_cp`) with ≥ 5 distinct cell lines. Replicate signatures within a drug–cell-line pair were averaged to yield one consensus signature per drug per cell line.

Genetic perturbation signatures were drawn from the same dataset by filtering to shRNA knockdown perturbagens (`pert_type = trt_sh`), yielding 154,993 signatures across 4,371 gene targets and 20 cell lines. Drug-target annotations from the Repurposing Hub mapped 1,067 drugs to gene targets present in the shRNA data.

Mechanism-of-action (MOA) annotations were obtained from the Broad Drug Repurposing Hub (version 2020-03-24) (Corsello et al., 2017), providing MOA labels for 1,782 of the 8,949 drugs. These labels were frozen before analysis (commit SHA `1dc20a2`).

2.2 Direction instability

For a drug tested in K cell lines, let $\mathbf{s}_1, \ldots, \mathbf{s}_K \in \mathbb{R}^{978}$ denote its consensus signature vectors. Define the unit-normalized signatures $\hat{\mathbf{s}}_i = \mathbf{s}_i / \|\mathbf{s}_i\|_2$. Direction instability is:

$$D = 1 - \frac{2}{K(K-1)} \sum_{i<j} \hat{\mathbf{s}}_i^\top \hat{\mathbf{s}}_j \tag{1}$$

$D \in [0, 2]$. $D = 0$ when all signatures point in the same direction. $D = 1$ when signatures are orthogonal on average. $D > 1$ indicates anticorrelation.

2.3 Complementary metrics

Magnitude CV: coefficient of variation of signature L_2 norms $\{\|\mathbf{s}_i\|\}_{i=1}^{K}$, capturing effect-size variation independent of direction.

Fréchet variance: mean squared geodesic deviation from the Fréchet mean on the unit sphere S^{977}.

Gene-level Jaccard overlap: for each pair of cell lines, the Jaccard index of the top-50 up-regulated and top-50 down-regulated genes, averaged over all pairs and both directions.

2.4 Population null model

A drug's direction instability depends on its cell-line count: drugs tested in more cell lines have more pairwise comparisons and tend toward higher baseline instability. To control for this confound, we compare each drug against a reference population of drugs with similar coverage: all drugs with cell-line count in $[K/2,\ 2K]$, where K is the focal drug's count. The p-value is the fraction of reference drugs with instability $\leq$ the focal drug's instability. This replaces a permutation null, which is invariant under cell-line label shuffling for pairwise cosine statistics.

2.5 Pre-registration

The scoring functions (Equation 1 and all complementary metrics) and the frozen MOA labels were committed before any real LINCS signatures were analyzed. The commit SHA and frozen label file are available in the repository.

3 Results

3.1 Landscape of drug mechanism transport

Among 8,949 drugs with ≥ 5 cell lines, mean direction instability was 0.924 ± 0.061 (median 0.940). Most drugs exhibit context-dependent effects. Only 78 drugs (0.9%) achieved $p < 0.01$ under the population null, and 411 (4.6%) achieved $p < 0.05$, indicating that genuinely mechanism-conserving drugs are rare.

Cell-line coverage ranged from 5 to 64 (mean 10.2, median 8). The population null accounts for this variation: a drug's instability is evaluated against drugs with comparable coverage.

3.2 Direction instability reflects biological mechanism conservation

Direction instability anticorrelated with gene-level Jaccard overlap of top perturbed genes: Spearman $\rho = -0.79$ ($p < 10^{-300}$). Drugs with low instability literally perturb the same genes across cell types, while high-instability drugs engage different gene programs in different contexts. This confirms that direction instability captures genuine biological mechanism change, not measurement noise.

3.3 Direction and magnitude instability are partially dissociable

Direction instability and magnitude CV showed weak correlation (Spearman $\rho = -0.10$, $p = 2.1 \times 10^{-21}$). 7.8% of drugs fell in the low-direction, high-magnitude quadrant: conserved mechanism with context-dependent penetrance.

3.4 MOA class stratification

Table 1 shows mean direction instability by MOA class (classes with ≥ 10 annotated drugs). Two classes stand out: HDAC inhibitors ($\bar{D} = 0.77$, $\sigma = 0.14$) and topoisomerase inhibitors ($\bar{D} = 0.77$, $\sigma = 0.12$). CDK inhibitors and tubulin polymerization inhibitors also show reduced instability ($\bar{D} = 0.80$ and 0.88 respectively). Receptor antagonists and agonists cluster near the population mean ($\bar{D} = 0.93$–0.96), consistent with receptor-mediated mechanisms being cell-type-dependent.

3.5 The HDAC inhibitor selectivity gradient

The HDAC inhibitor class provides a natural experiment: 20 annotated HDAC inhibitors span the full selectivity spectrum from pan-isoform to isoform-specific. Table 2 shows the eight most-transporting HDAC inhibitors (all pan-isoform) and the five least-transporting (selective or weak).

Pan-HDAC inhibitors (mean $D = 0.62$, $n = 8$) transport their mechanism across cell lines because their targets—histone deacetylases of classes I, II, and IV—are expressed in every cell type. Chromatin remodeling is a universal process, so inhibiting it broadly produces a conserved transcriptomic signature.

Selective HDAC inhibitors (mean $D = 0.96$, $n = 5$) do not transport. PCI-34051 targets HDAC8 specifically; its transcriptomic effect depends on whether HDAC8 is functionally relevant in a given cell type. Valproic acid and phenylbutyrate are weak HDAC inhibitors with additional pharmacological activities, so their signatures reflect cell-type-specific off-target effects rather than HDAC inhibition per se.

Entinostat ($D = 0.70$, class I–selective) falls between the two groups, consistent with intermediate target breadth: class I HDACs (HDACs 1, 2, 3) are more widely expressed than HDAC8 but narrower than the full isoform panel.

3.6 Top transporting drugs target fundamental machinery

The 10 most mechanism-conserving drugs with MOA annotations target processes common to all cells (Table 3): protein synthesis (homoharringtonine), RNA transcription (dactinomycin), DNA topology (epirubicin, doxorubicin), chromatin remodeling (PCI-24781, belinostat, panobinostat), cell cycle progression

Table 1: Direction instability by MOA class ($n \geq 10$). Classes targeting fundamental cellular machinery (chromatin, DNA, cell cycle) show lower instability than receptor-mediated classes. Within HDAC inhibitors, the high variance ($\sigma = 0.14$) reflects the pan-selective gradient discussed in Section 3.5.

MOA class	n	$\bar{D}$	σ
topoisomerase inhibitor	17	0.765	0.123
HDAC inhibitor	20	0.766	0.142
CDK inhibitor	16	0.797	0.119
tubulin polymerization inhibitor	19	0.882	0.048
MEK inhibitor	14	0.888	0.066
EGFR inhibitor	23	0.899	0.064
FLT3 inhibitor	15	0.896	0.057
tyrosine kinase inhibitor	14	0.901	0.065
VEGFR inhibitor	21	0.909	0.045
calcium channel blocker	25	0.923	0.067
monoamine oxidase inhibitor	17	0.924	0.071
glucocorticoid receptor agonist	31	0.933	0.039
Population mean	8,949	0.924	0.061
dopamine receptor antagonist	59	0.936	0.046
histamine receptor antagonist	41	0.944	0.049
serotonin receptor antagonist	62	0.945	0.035
cyclooxygenase inhibitor	53	0.947	0.031
adrenergic receptor antagonist	52	0.945	0.038
adrenergic receptor agonist	40	0.950	0.028
phosphodiesterase inhibitor	31	0.953	0.023
serotonin receptor agonist	31	0.958	0.022
PPAR receptor agonist	19	0.956	0.019

(BMS-387032), and signal transduction kinases (bisindolylmaleimide-IX). These mechanisms operate independently of tissue-specific receptor expression, explaining their cross-context conservation.

Conversely, imatinib ($D = 0.97$) exemplifies a context-dependent drug: its BCR-ABL kinase inhibition produces strong effects only in cells harboring the Philadelphia chromosome, yielding high direction instability across the LINCS panel.

3.7 Direction instability is not driven by cytotoxicity

A potential confound: drugs that kill cells uniformly might produce conserved signatures through a generic stress response rather than target-specific mechanism transport. We tested this by computing the mean cytotoxic signature across known cytotoxic drugs (topoisomerase inhibitors, DNA alkylating agents, tubulin inhibitors, proteasome inhibitors, RNA polymerase inhibitors, protein synthesis inhibitors) and measuring each drug's cosine similarity with this reference.

Direction instability showed weak correlation with cytotoxic-signature overlap (Spearman $\rho = -0.17$, $p = 1.6 \times 10^{-58}$, excluding known cytotoxic drugs). The correlation is in the wrong direction for the confound hypothesis: drugs resembling the cytotoxic profile are *less* stable, not more. Removing the 50 genes most associated with the cytotoxic signature and recomputing direction instability preserved the HDAC inhibitor gradient (Spearman $\rho = 0.83$ between original and stress-corrected values). Pan-HDAC inhibitors became

Table 2: HDAC inhibitors ranked by direction instability. Pan-isoform inhibitors achieve low instability and significant population p-values; selective inhibitors are indistinguishable from the null.

Drug	Selectivity	D	n	p
PCI-24781	pan	0.561	13	0.004
belinostat	pan	0.589	13	0.006
panobinostat	pan	0.591	15	0.007
scriptaid	pan	0.620	15	0.009
vorinostat	pan	0.628	60	0.008
trichostatin-a	pan	0.634	64	0.014
NSC-3852	pan	0.674	15	0.016
entinostat	class I	0.697	14	0.023
valproic acid	weak/partial	0.955	54	0.947
Merck60	selective	0.962	9	0.778
PCI-34051	HDAC8-selective	0.984	12	0.976
phenylbutyrate	weak/partial	0.989	11	0.988

Table 3: Top 10 transporting drugs with MOA annotation.

Drug	MOA	D	p
homoharringtonine	protein synthesis inhibitor	0.497	0.001
dactinomycin	RNA polymerase inhibitor	0.506	0.002
BMS-387032	CDK / cell cycle inhibitor	0.522	0.002
bisindolylmaleimide-IX	PKC inhibitor	0.526	0.003
epirubicin	topoisomerase inhibitor	0.529	0.003
anisomycin	DNA synthesis inhibitor	0.555	0.004
PCI-24781	HDAC inhibitor	0.561	0.004
BNTX	opioid receptor antagonist	0.564	0.003
digitoxigenin	ATPase inhibitor	0.586	0.004
belinostat	HDAC inhibitor	0.589	0.006

more stable after stress-gene removal (vorinostat: $D = 0.63 \to 0.50$), confirming that their conserved signal is chromatin-specific, not a generic stress response.

3.8 The transporting core is the target pathway

For each drug, we defined a "transporting core": genes with consistent sign (>80% of cell lines agree), low magnitude CV (<1.0), and meaningful effect size (mean $|z| > 0.5$). Core size anticorrelated with direction instability (Spearman $\rho = 0.69$, $p < 10^{-300}$), confirming that transporting drugs affect a larger set of genes consistently.

The 8 pan-HDAC inhibitors shared a 26-gene core—genes consistently perturbed by all 8 drugs across all cell lines. This shared core includes *SUV39H1* (a histone methyltransferase directly involved in chromatin remodeling), *MYC* (a master transcription factor regulated by HDAC activity), *CDK6* (cell cycle kinase transcriptionally repressed by HDAC inhibitors), *BIRC5* (survivin, a known HDAC-responsive anti-apoptotic gene), and *ORC1* (DNA replication origin licensing). The transporting core is the target pathway: pan-HDAC inhibitors share a chromatin/cell-cycle gene program that operates identically across cell types.

Table 4: Exploratory within-family genetic triangulation: Spearman correlation between direction instability and |drug–shRNA cosine| for each drug's annotated target. Negative ρ indicates that mechanism-conserving drugs (low D) produce signatures more similar to genetic knockdown of their target. This stratification was not pre-registered. Families shown have ≥ 15 drug–target pairs and $p < 0.10$.

MOA family	n	ρ	p
PI3K inhibitor	18	−0.55	0.017
adrenergic receptor agonist	50	−0.45	0.001
topoisomerase inhibitor	18	−0.48	0.045
EGFR inhibitor	27	−0.40	0.038
HDAC inhibitor	93	−0.32	0.002
Global (all families pooled)	2,138	−0.18	1.1×10^{-16}

3.9 Genetic triangulation: drug transport tracks target knockdown

The preceding analyses establish that direction instability tracks mechanism conservation, but they rely on drug signatures alone. We tested a stronger prediction: if a transporting drug genuinely acts through its annotated target, its perturbation signature should resemble the transcriptomic effect of genetically knocking down that target. We computed cosine similarity between each drug's consensus signature and the shRNA knockdown signature of its annotated target gene, matched within the same cell line.

Among 2,138 drug–target pairs with ≥ 3 shared cell lines, the pre-registered global correlation between direction instability and |drug–shRNA cosine| was significant (Spearman $\rho = -0.18$, $p = 1.1 \times 10^{-16}$) and in the expected direction, but fell below our pre-registered effect-size threshold of $|\rho| > 0.3$.

A structural limitation of the pooled test is that it conflates drug families with inherently different baseline drug–knockdown cosines. Proteasome inhibitors produce signatures that naturally overlap with proteasome-subunit knockdown (cos $\approx$ 0.45), while HDAC inhibitors produce lower cosines with individual HDAC knockdowns (cos $\approx$ 0.15), irrespective of instability. Pooling across families introduces variance that is orthogonal to the instability signal.

We therefore performed an exploratory within-family stratification, which was not pre-registered (Table 4). Five families exceeded $|\rho| > 0.3$ at $p < 0.05$, spanning kinase inhibitors, chromatin modifiers, and receptor agonists. The expected sign held in all five: lower instability predicted higher drug–target cosine. Per-target-gene analysis showed convergent effects: KDR (VEGFR, $\rho = -0.84$, $p = 5.1 \times 10^{-7}$), CDK1 ($\rho = -0.79$, $p = 0.002$), AKT1 ($\rho = -0.75$, $p = 0.002$), and HDAC6 ($\rho = -0.71$, $p = 0.009$). These within-family results are exploratory and require independent replication.

The HDAC family again provides the most granular view. Among 100 HDAC drug–target pairs, pan-isoform inhibitors (PCI-24781, belinostat, panobinostat, vorinostat) showed mean |cosine| of 0.12–0.18 against their HDAC targets. PCI-34051 (HDAC8-selective, $D = 0.98$) achieved only 0.037 cosine with HDAC8—its putative primary target—and 0.10 against off-target HDACs. The selectivity gradient that appeared in the direction-instability ranking reappears in the genetic triangulation: pan-inhibitors produce transcriptomic effects that recapitulate target knockdown; selective inhibitors do not.

3.10 Pre-registered predictions: hits and misses

We pre-registered the invariance-graph analysis before running it: for each drug, we built a graph over cell lines (edge = cosine similarity ≥ 0.5) and examined connected components. We predicted that pan-HDAC inhibitors would form 1–2 components while selective inhibitors would fragment completely. The ordinal gradient held: vorinostat formed 1 component across all 60 cell lines; PCI-34051, tubastatin-a, phenylbutyrate, and JNJ-26481585 each fragmented into $K = n$ components (every cell line isolated). The strict threshold did not hold: pan-HDAC inhibitors formed 1–5 components rather than the predicted 1–2 (e.g., PCI-24781 had 4 components across 13 cell lines). The gradient is unmistakable; the pre-registered threshold was too strict.

We also predicted that connected components would correspond to tissue lineages. Only 1 of 163 drugs with 2–5 components showed significant tissue enrichment in any component (Fisher's exact test, $p < 0.05$), a clear miss on the pre-registered prediction. This null is itself informative: invariance-graph structure reflects mechanism-level conservation, not tissue-of-origin expression clustering. Drugs that transport do so because of their target pathway, not because of shared lineage among responding cell lines.

4 Discussion

Direction instability provides a single-number summary of whether a drug's transcriptomic mechanism generalizes across cell types. Five findings support its validity. The metric tracks gene-level mechanism conservation ($\rho = -0.79$ with Jaccard overlap). MOA classes stratify in a biologically interpretable pattern, with machinery-targeting drugs showing low instability and receptor-mediated drugs showing high instability. The HDAC selectivity gradient provides a within-class controlled comparison that rules out MOA labels as the driver. The cytotoxicity defense confirms that the signal reflects target-specific mechanism conservation, not generic cell death.

The transporting-core analysis identifies the specific genes that are conserved: for pan-HDAC inhibitors, the shared core is the HDAC target pathway itself (*SUV39H1*, *MYC*, *CDK6*). Genetic triangulation provides suggestive convergent evidence from an independent data modality: within five drug families, exploratory within-family analysis found that mechanism-conserving drugs produce signatures more similar to genetic knockdown of their annotated targets, though the pre-registered global test fell short of its effect-size threshold. Independent replication of the within-family result would strengthen the case that transport is target-mediated.

Relation to the bracket norm. Direction instability is formally analogous to the neural bracket norm (Tower, 2026), which measures directional rotation of learned representations across evidence levels. Both metrics detect when a system responds to a perturbation by rotating its internal state rather than scaling it. The pharmacological application replaces evidence quartiles with cell lines and neural activations with gene expression signatures, preserving the geometric interpretation: low rotation implies a conserved mechanism that transports across contexts.

Honest nulls. Three pre-registered predictions missed their thresholds. The global genetic triangulation correlation ($\rho = -0.18$) fell below the pre-registered bar of $|\rho| > 0.3$; the within-family stratification that rescued this result was not pre-registered. Component-tissue enrichment showed no signal (0.6% vs. the predicted >20%). The strict component-count prediction for pan-HDAC inhibitors (1–2 components) was too tight (observed 1–5). All three are reported because the pre-registration commit SHA proves they were genuine predictions, not post-hoc hypotheses.

Limitations. LINCS L1000 measures only 978 landmark genes; full-transcriptome profiling might reveal additional structure. The population null controls for cell-line count but not for other confounds (drug concentration, time point, signature quality). MOA annotations from the Repurposing Hub are incomplete (20% coverage) and may contain errors. The direction-instability metric treats all cell lines as exchangeable and does not model phylogenetic relationships between cell types.

Applications. Direction instability could serve as a filter for identifying compounds whose transcriptomic mechanism is conserved across cellular contexts, a prerequisite for robust in-vitro-to-in-vitro translation. Whether cell-line transport predicts clinical efficacy across patient populations remains an open question outside the scope of this work. The metric requires only perturbation signatures across ≥ 5 contexts, making it applicable to any multi-context profiling dataset (CMAP, JUMP-CP, Perturb-seq).

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